

Tunisia — the complete program. Everything, in one document.

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt \$40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for \$680M), strategic materials (phosphogypsum at Gabès costing the sea >\$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	gross export	716,500 lb U3O8 + 585 t REO	\$80-95/lb U3O8 · \$30/kg REO	\$75-86M	STRONG
Solar fuel substitution (500 MW)	import substitution	745 GWh/yr displaced (876 GWh less 131 GWh to H2)	\$142/MWh (gas \$12/MMBtu x 11 MMBtu/MWh + \$10/MWh O&M); government anchor: 1,100 GWh ≈ \$125M/yr gas saving for the same 500 MW fleet (Pg14)	\$95-116M	PLAUSIBLE
Hydrogen → ammonia (loop-internal offtake)	import substitution	2.63 kt H2/yr → 14.9 kt NH3 for the GCT chain	\$500/t imported ammonia displaced (external offtake upside: \$8-9.5/kg H2)	\$7-25M	SPECULATIVE
Terfas (desert truffles)	gross export	940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported)	EUR20-40/kg farm-gate	\$14-29M	PLAUSIBLE

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
Food import substitution	import substitution	1% of the \$3.0B food import bill (frame; volume bridge is a pilot deliverable)	imported protein \$15-25/kg vs local \$5-10/kg	\$30-30M	<i>SPECULATIVE</i>
Compute colocation services	gross export (services)	colo park, tenant-dependent	market colocation rates	\$15-15M	<i>SPECULATIVE</i>
ESG financing delta	financing delta (not FX)	100-150bp on ~\$800M portfolio debt	interest saved, not foreign exchange	\$8-12M	PLAUSIBLE
Total external-account effect (mid)				~\$279M/yr	\$80M strong · \$137M plausible · \$61M speculative · \$0 verified

By kind: exports \$117M · import substitution \$152M · financing delta \$10M (interest saved, honestly labeled as not-FX). **By grade:** \$80M strong · \$137M plausible · \$61M speculative · \$0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

3. The energy ledger — every kWh, by leg

LEG	GWH/YR	THE FORMULA
Pump-as-turbines (return flows)	3.9	$70\% \times 150\text{k m}^3/\text{d} \times 50\text{ m} \times 75\% \times 8,760\text{ h}$
Heat (kiln + exotherm)	30	9% of 333 GWh envelope
Flexibility (load shift)	20	dispatch value — not recovered joules
CO ₂ ; mineralized	30 kt	mass, not energy
Total	33.9 recovered + 20 dispatch	isobaric ERD inside the 4 kWh/m³ benchmark — never double-counted

4. Phase 0 — every dollar, itemized

ITEM	COST	WHAT IT BUYS
Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)	\$35,000	the first anchor measurement
Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access	\$40,000	the second anchor measurement
Emissions inventory: 30 passive samplers (PM _{2.5} /SO ₂ /fluorides) + lab	\$25,000	baseline for the health scoreboard

ITEM	COST	WHAT IT BUYS
Cohort protocol: 3,000-family registry, ethics, data platform v0	\$35,000	the DALY ledger becomes MEASURED
Health pilot first tranche: 100 households + 2 schools, 90 days	\$58,000	filters, clean-air boxes, KPIs
Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration	\$25,000	structure before scale
Engineering + project management	\$35,000	the 90 days are run
List price	\$253,000	list price \$253k; the two anchor measurements cost \$75k and land first — the claimed range is the phased-execution plan, not the list

5. Every claim, graded

CLAIM	GRADE	WHY
\$40.5B debt (~79% GDP); 26.2% of export revenues to service	STRONG	MEASURED via World Bank API on 2026-09-09: debt \$40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)
\$3.0B food and \$4.9B fuel imports	STRONG	fuel \$4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification
InstaDeep founded in Tunis, sold for \$680M	STRONG	public deal record; the emblem for talent without infrastructure
Gabès: >\$58M/yr of REE value lost to the sea	STRONG	Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~\$58M/yr losses
U + heavy REE from the acid stream, ~\$80M/yr	STRONG	flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0
~\$285M/yr external-account effect, tiered	TIERED	\$80M strong (proven flowsheet/cited anchor) · \$137M plausible (gated) · \$68M speculative (assumed volume/price)
33.9 GWh/yr recovered + 20 GWh/yr dispatch value	DERIVED	every leg computed from cited primitives; dispatch value honestly labeled — not joules
\$2,592 per healthy year (water rung)	PLAUSIBLE	arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED
Phase 0 = \$150-250k, itemized	PLAUSIBLE	itemized list price \$253k; the two anchor measurements cost \$75k and land first
Commercially validated	ZERO	no buyer has signed anything — published as zero on purpose

6. The hidden dependencies

DEPENDENCY	TYPE	STATUS	NOTE
Desal brine permit	regulatory	in design	Gabes outfall / diffuser; permit required before SWRO FID
PG chemistry (Ra-226 partitioning)	technical	Phase 0 assay	flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	regulatory	gate	CNSTN licensing, NORM transport, export controls
REE separation capex	capital	later phase	+\$244M solvent-extraction plant beyond the ~\$100M circuit
Solar bankability	financial	gate	DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	infrastructure	later phase	3 GW connection + 400 kV upgrades ~\$450M
Compute accelerators	regulatory	gate	export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	agronomic	gate	376 kg/ha is the 20-yr Murcia average; gate = 3-season record ≥ 200 kg/ha
Export prices	market	exposure	U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	financial	uncommitted	DFI channel modeled for solar/desal; nothing committed

7. The structure and the pilot/capital separation

The design is six independently bankable modules sharing infrastructure. The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDD (2023): Tunisia's water-sector investment need ~\$17.1B and energy-sector ~\$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is \$150-250k of measurements, and the whole core build is ~\$250-650M phased over years 2-5, module by module, each gated. The CCDD numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~\$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~\$31/MWh awards are the reference the DSCR gate is tested against.

8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B, 78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous,

non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop’s answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

The household ledger — what this does to the cost of living

A state program that does not make life cheaper for its people is decoration. Each line, computed from the same primitives as the rest of this document; the quarterly price index in Gafsa and Gabès is the published enforcement: **the price of protein and vegetables must bend downward within 24 months of the first harvest.**

HOUSEHOLD LINE	TODAY	AFTER THE LOOP
Water	\$1.00-1.50/m ³ (SONEDE produced)	\$0.35/m ³ coastal · \$0.34/m ³ inland
Protein	\$15-25/kg imported	\$5-10/kg local
Terfas	export €20-40/kg	one third stays home at ≤€12/kg, contractual
Health	the disease's bill	\$2,592 per healthy year vs the \$4,000/DALY standard
Energy (grid)	\$0.110/kWh industrial	\$0.031/kWh for the loop's loads; tariffs stay state policy
Work	28% unemployment (54% women graduates)	workshop, nursery, ponds, plants, crews

The energy ledger — what power costs, per party

Anchors cross-checked: Rw10 (0.352 TND/kWh industrial, 19 TWh), Pg14 (the government’s own 1,100 GWh = 5% of production, ~\$125M/yr gas saving), Rc13 (\$31/MWh inside the \$24-35 historical bid band), Pg15 (fuel imports \$4.895B, WITS-verified).

PARTY	TODAY	AFTER THE LOOP	THE DELTA
The loop's industries	grid \$0.110/kWh	\$0.031/kWh on the loop's PPA	−72%
The state	19-22 TWh/yr, ~95% gas, \$4.90B fuel bill	876 GWh/yr solar (4.0-4.6% of production)	−\$106-125M/yr gas imports (\$114-142/MWh band)
The household	subsidized STEG tariff	the cost stack beneath the tariff falls	the inflation channel throttled — every dinar protected

The frontier — where the loop reaches, and where it honestly stops

The complete household budget, swept — nothing hidden, nothing promised beyond the frontier.

HOUSEHOLD LINE	STATUS	THE HONEST NOTE
Electricity (household tariff)	indirect	household tariffs stay subsidized STEG policy; the loop lowers the cost stack they sit on and the subsidy burden the state carries
Cooking gas (the bonbonne)	out of scope	the loop displaces grid gas, not bottled LPG; the bonbonne subsidy is a separate state decision
Transport fuel	out of scope today	the hydrogen leg serves ammonia for the fertilizer chain; transport fuels are a later policy question
Wheat and bread subsidy	out of scope	the protein stack complements the grain program; it does not replace it
Water	covered	cost floor \$1.00-1.50 -> \$0.35 coastal / \$0.34 inland
Health	covered	\$2,592 per healthy year vs the \$4,000 standard
Protein	covered	\$5-10/kg local vs \$15-25/kg imported
Work	covered	the workshop, nursery, ponds, plants, crews where unemployment is 28% (54% for women graduates)
Housing	out of scope	the loop does not build housing; named so it is not promised

The water stack — line-by-line, with each credit's failure mode

The \$0.347/m³ audit: the base is measured anchors; each credit is a loop-internal yield with a named failure mode; the degraded cascade still clears the \$0.68 gate.

LINE	\$/M ³	PROVENANCE	WHAT MAKES IT FAIL
Base — solar energy 4 kWh/m ³ at the loop's own PPA + capex + O&M	\$0.567	MEASURED anchors (Rc13/Rw2/Rw3)	holds — if even the PPA lapses, grid power takes the base to \$0.88, still under SONEDE
DC reject heat → RO feed preheat — one seawater pipe cools the servers, then arrives warm at the membranes (SEC −2.5%/°C)	−\$0.007	DERIVED — Leg 03 + Leg 01, same pipework	fails only if no compute anchor tenant — and costs \$0.007
Kiln/exotherm preheat — the acid plant and board kilns warm the same feed	−\$0.000	DERIVED — Leg 04/05 waste heat	negligible by design — carried at zero margin
Brine valorization — salt/Mg evaporated on the loop's own free heat, sold at the park gate (of-take signed; nothing hauled)	−\$0.137	DERIVED — brine is the desal's own by-product	if the of-take is not signed by FID, the credit degrades to −\$0.03 (disposal avoided only) — the biggest single sensitivity
Shared civil works — one intake, one outfall, one grid connection serves RO + DC + metals in the same park	−\$0.032	DERIVED — 12% capex sharing	if the park phases one module at a time, halve it to −\$0.016

LINE	\$/M ³	PROVENANCE	WHAT MAKES IT FAIL
Dispatch value — the desal's own flexibility follows the loop's own solar curve	−\$0.015	DERIVED — 20 GWh shifted	fails to zero if the STEG contract does not reward flexibility — contractual, not hoped
ESG financing delta — the health leg's public KPIs reprice the loop's own debt (100–150bp)	−\$0.026	DERIVED — nations_v3	fails to zero if lenders do not accept the KPI track record — bankable-region fallback keeps the module alive
Brine pump-as-turbine — the desal's own residual pressure	−\$0.002	DERIVED — 2.5 bar × 75%	fails to zero — immaterial
Integrated	\$0.347/m³	28% of SONEDE's \$1.00-1.50 (Pg16)	the degraded cascade: if every soft credit fails, water still lands at \$0.521/m³ — under SONEDE's own \$1.00–1.50, and the \$0.68 gate still clears with margin. The credits are the margin, not the plan.

The Darwinian ledger — why business as usual loses

Protein built the human brain (Pg27); the same lever, applied on purpose, is the plan's deepest one. The business-as-usual ledger is the fear-facts: what “keep doing what we do” actually buys.

FACT	THE NUMBER	WHAT IT MEANS FOR THE NEXT GENERATION'S BRAINS
Fluorosis belt	24% dental fluorosis (Rw1)	high-F water ↔ lower child IQ (Pg28) — BAU taxes the cortex before it forms
Protein poverty	first 1,000 days (Pg29)	permanent cognitive deficit — the trees version, built child by child
Brain drain	tertiary emigration among world's highest (Pg30)	the country exports its cortex like its phosphate
The proof	InstaDeep sold for \$680M	the cognitive gold was real — and sold, not kept
Debt	26.2% of export revenues to service (Pg13)	the budget cannot buy the interventions the brain needs
The idle cohort	28% unemployment; 54% women graduates (Rm19)	built brains, no environment — adaptation = exit
Inflation	5.15% on \$4.90B fuel + \$3.00B food	the first cut is diet quality — the Fed reaches the children's IQ
SELECTION PRESSURE	WITHOUT THE PLAN	WITH THE PLAN
Water scarcity	selects emigration — the skilled leave first	the mesh re-specifies the environment; adaptation becomes staying
Protein poverty	caps cognition permanently	ponds + canteens + quota reverse the pressure
The debt cycle	the surplus is eaten; the economy cannot invest	owned exports + substitution cut the invoice
Institutions	gates stay closed; skills emigrate through them	public gates select for competence, in public

The open-field plan — terfas, hectare by hectare

RUNG	HECTARES	SEEDLINGS	PRODUCTION	VALUE	OPENS ON
Gate (yrs 1-4)	100	600,000	≥200 kg/ha	first fruit ≥€20/kg	3-season record + contract
Scale (yrs 4-8)	2,500	15M	940 t/yr	€16544-29704M/yr	nursery output
Ceiling (yrs 8+)	3,000	18M	1,128 t/yr	inside 1,000-1,500 t/yr market cap (Rc8)	market offtake signed

6,000 mycorrhized seedlings/ha (Rc16) — the nursery is the first asset: 5M seedlings/yr of local-ecotype capacity, ~200 nursery + ~1,000 field + ~783 seasonal jobs at scale. Local wild genetics, not the imported soil biology that failed in Saudi Arabia (Rc17).

The AI spine — the system's nervous system, and its boundary

The AI runs the measurement, the routing and the publication — never the decisions.

COMPONENT	WHAT IT DOES	WHAT IT IS WORTH
The dispatcher	routes every joule and m ³ in real time: solar → desal → electrolyzer → ponds → DC cooling, following the solar curve minute by minute	worth the 20 GWh/yr of dispatch value — the AI is what collects it
The scoreboard	computes and publishes every gate number automatically — filter uptime, urinary F, DSCR, yield records — tamper-evident, no human can edit a gate	the public contract, machine-audited
The price index	tracks the Gafsa/Gabès household basket quarterly and publishes the bend — the people's enforcement metric, computed not curated	the number that proves the prices bend, or proves they do not
The cohort engine	runs the 3,000-family health baseline, dose-outcome analytics and anomaly flags for the clinical team	the DALY ledger becomes a live dataset
The dispatcher twin	a digital twin of the loop (energy, water, brine, heat) that simulates before any physical change and audits after	the falsification ladder, automated
The boundary	the AI never sets a gate, never signs a contract, never prices a tariff — it measures, routes and publishes. The decisions stay human, in public.	the guardrail that keeps the nervous system a servant, not a master

The results framework — what a World Bank board reads first

The scoreboard IS the M&E system (PforR class, Pg23).

INDICATOR	BASELINE	TARGET	DATA SOURCE
Filter uptime, pilot households	0% (filters not installed)	≥70%	scoreboard, live
Urinary fluoride, pilot children	baseline to be measured (Day 1-30)	–20%	cohort engine

INDICATOR	BASELINE	TARGET	DATA SOURCE
Household price index, Gafsa/Gabès	baseline Qo	bend downward ≤24 months after first harvest	price index, quarterly
Water produced cost	SONEDE \$1.00-1.50/m ³ (Pg16)	≤\$0.35/m ³ integrated	plant telemetry
Energy recovered inside the loop	o GWh/yr	33.9 GWh/yr + 20 GWh dispatch	dispatcher ledger
Gas imports displaced	\$4.895B fuel bill (Pg15)	–\$106-125M/yr	WITS Comtrade, annual
Uranium recovered from the acid stream	o (lost to the sea)	≥85% recovery gate	assay chain
Terfas hectares planted	o ha	100 ha gate → 2,500-3,000 ha	nursery registry
Jobs in the basin	28% unemployment (54% women graduates)	workshop → nursery → plants → crews	employment registry

The people's energy is electrons, not hydrogen — and the H₂ lives inside the loop

The loop's H₂ (2.63 kt/yr) is an industrial molecule: per kWh of heat, green H₂ at \$0.21 is 7× the loop's \$0.031/kWh electron; blending caps at 5-20% by volume (Pg20/Pg21/Pg22); the whole H₂ volume equals 0.0% of grid-gas demand. The molecule goes where only a molecule works: ammonia for the GCT chain, inside the loop. the electrolyzer sits on the desal line, and everything it touches stays in the loop: the water it splits is the SWRO permeate (0.04% of the 150,000 m³/d output); its un-split reject stream is clean water that returns to the product line; its oxygen by-product aerates the spirulina ponds and greenhouses; its stack heat (15–20% of input) joins the kiln and DC reject heat on the desal preheat circuit; and the minerals concentrated in its flush stream join the brine valorization line. The electrolyzer is not a consumer of the loop — it is another recovery point inside it.

The total routing — every stream gets a home

The ponds do *not* drink brine: SWRO brine is 55-70 g/L, past Arthrospira's ~30 g/L ceiling (Pg31); they drink the desal permeate and breathe the electrolyzer's oxygen. The full inventory:

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Solar electrons	500 MW fleet	desal, metals, DCs, electrolyzer, pumps — routed by the dispatcher	876 GWh/yr
SWRO permeate	coastal desal	city offtake · electrolyzer feed · ponds · greenhouses	150,000 m ³ /d
SWRO brine	coastal desal	salt + magnesium valorization, evaporated on the loop's own free heat	122,727 m ³ /d → 58 kt Mg/yr
Brine bittern	valorization	Mg(OH) ₂ ; fire retardant → the board's own chemistry	140 kt/yr = 140% of the board's need

STREAM	PRODUCED BY	ITS HOME	THE NUMBER
Electrolyzer oxygen	H&sb2; plant	spirulina pond + greenhouse aeration	21 kt/yr
Electrolyzer hydrogen	H&sb2; plant	ammonia → the GCT fertilizer chain	2.63 kt/yr
Electrolyzer reject water / heat / flush	H&sb2; plant	product line / desal preheat / brine line	nothing leaves
DC reject heat	compute park	the same seawater pipe, arriving warm at the membranes	−\$0.007/m ³
Kiln + acid exotherm	board kilns + metals plant	desal preheat + greenhouses	30 GWh/yr
CO&sb2; (kiln + power)	combustion streams	mineralized into board (2 Mt PG) · pond carbon	540 t CO&sb2;/yr to the ponds
Phosphogypsum	GCT stack	fire-rated board — with its retardant from our own brine	2 Mt/yr
Acid stream	GCT wet-process acid	U + heavy REE extraction; raffinate back to fertilizer	716.5k lbs U&sb3;O&sb8;/yr
Municipal return flows	city mesh	30%+ reuse · pump-as-turbines on the descent	3.9 GWh/yr
Sludge / digestate	wastewater + organics	biogas for kilns and greenhouses	booked option, honestly labeled
Slaughterhouse bones	local slaughterhouses	bone char → the fluoride filters; spent char → phosphate soil amendment for the fields	filter loop closes into fertilizer
Spent pond medium	spirulina harvest	nutrient-rich → greenhouse fertigation	nothing discarded
Ammonia fertilizer	GCT chain	the terfas fields' host plants + greenhouse soils	the truffle grows on the loop's own nitrogen
Compute services	colo park	export — the data product	tenant-gated
Ra-226 / Po-210	acid stream radiology	stays in the gypsum/board matrix — the radiology gate	below the 1,000 Bq/kg exemption
Board offcuts	board line	recycled into new board	zero waste
Sodium chloride	brine line	industrial offtake at the park gate	nothing hauled
Halophytes / Artemia on brine	brine line	future option, honestly labeled — not a current credit	the only stream awaiting its consumer

the ponds drink the desal permeate — ~{:.0f} m³/d, {:.2f}% of the 150,000 m³/d output — and breathe the electrolyzer's oxygen; the brine (55–70 g/L, far past the ~30 g/L ceiling the organism tolerates) is routed to salt and magnesium valorization instead. Halophytes and Artemia on brine are booked as a future option, honestly labeled — not a current credit.

9. The state program — architecture, instruments, financing, adoption

Counterparts — who signs what

INSTITUTION	INSTRUMENT	ROLE
GCT (Tunisian Chemical Group)	acid-stream MOU + sampling agreement	the treasury's feedstock
SONEDE	municipal/industrial water offtake at $\geq \$0.68/\text{m}^3$ gate	water contracts under the existing regime
STEG	25-year PPA under the IPP regime (500 MW class)	the power leg's revenue
Ministry of Industry, Mines & Energy	program sponsor; tender registration	the executive mandate
Ministry of Health	biomonitoring + cohort registry	the scoreboard
CNSTN	NORM licensing for the uranium route	the regulatory pass
Ministry of State Domains & Agriculture	40-year leases under Law 93-120 (86,000 ha available)	the land
BCT	reserve/rating path; FX policy coordination	the monetary endgame

Legal instruments — nothing new needs to be invented

INSTRUMENT	WHAT IT PROVIDES	CITATION
Organic Law 2016-35 (BCT)	monetary autonomy; the program does NOT monetize	Pg4/Pg5
IPP regime — Law 2015-62 & 2024/25 awards	renewables concessions; $\sim \$31/\text{MWh}$ PPAs; 500 MW precedent	Rc13
PPP framework — Law 2015-49	desal/metals as PPPs where the state is the offtaker	Pg2
Law 93-120 (state-domain leases)	40-year agricultural leases; $\sim 86,000$ ha earmarked	Rc19
NORM/radiological regime (CNSTN, IAEA standards)	the uranium route's license layer	Pg11/Rm12

Financing stack — staged, gated, modeled in code

STAGE	FINANCING	GATE / CONDITION
Phase 0 (\$150-253k)	grant / founder / early mandate	no DFI needed; 90 days
Health pilot (\$58k)	ESG/grant	the first scoreboard numbers
Metals plant ($\sim \$100\text{M}$)	60% debt @7% + offtake security; MIGA	DSCR ≥ 1.2 at contract prices (nations_v5)
Solar (500 MW, $\sim \$550\text{M}$)	DFI stack: EIB/AfDB/WB-IFC at 4.5% + MIGA; or lean build	DSCR gate inside the bankable region (nations_v5)
Desal ($\sim \$165\text{M}$)	70% debt @6% + municipal offtake + sovereign backstop	blended $\geq \$0.68/\text{m}^3$, $\geq 60\%$ pre-sold
Terfas + food (\$7-35M)	concessional agri + founder	nursery first
EU Global Gateway (17 GW)	the national envelope this program plugs into	Pg7

Governance

- every gate is a public number — published, dated, re-checkable
- health fund separate from the industrial SPVs; one shared data platform
- quarterly scoreboard: water chemistry, filter uptime, cohort KPIs, price index in Gafsa/Gabes
- each pilot passes or dies alone, in public — the falsification ladder is the governance
- commercially-validated ledger published at zero until the first signature moves it

Adoption path — what the state signs, in order

1. **Month 0:** the state signs: GCT sampling MOU + Phase 0 mandate (\$150-250k, 90 days, no construction)
2. **Day 90:** two anchor measurements land: the 100-point water panel and the acid-stream assays — every projection becomes a signed number
3. **Months 4-18:** pilots on public gates: acid pilot, 100 ha terfas + nursery, spirulina pond, solar bid, desal offtake, health KPIs
4. **Years 2-5:** build only what passed: metals, solar, desal, health at basin scale — module by module, each with its own financing
5. **Years 5-10+:** the full loop: municipal spine, compute colocation, hydrogen, brine valorization — the destination the original design described

Investor essentials — the FX bridge tiered, and the pathway economics

By grade: **\$80M strong** · \$137M plausible · \$61M speculative · \$0 validated — the zero is published on purpose, and each gate is the instrument that moves it. Pathway economics, computed in code (nations_v5.py):

PATHWAY	REVENUE	CAPEX	GROSS MARGIN	PAYBACK	DSCR
P2 metals	\$75-86M/yr	~\$100M	0-37%	6.3 yr (mid)	2.40 @60% debt
P3 solar	\$27.2M/yr	~\$0M	71%	n/a	0.46 mkt / 0.69 DFI — clearing PPA ≥\$45652/MWh
P4 desal	\$30.1M @\$0.55 → \$38.3M @\$0.70	~\$0M	gate-dependent	n/a	0.54 @\$0.55 → 1.36 @\$0.70 — clears

Engineer essentials — the numbers an EPC checks first

Energy: 33.9 GWh/yr recovered + 20 GWh dispatch — PAT 3.9, heat 30, CO₂ 30 kt mineralized.
Water: SWRO 4 kWh/m³ lift floor 1.09 kWh/m³ per 400 m (Rw4); gate \$0.68 blended, ≥60% pre-sold.
Materials: Gafsa rock 322 mg/kg REE avg (Rm1); 60-85% to PG (Rm2); 85-90% of rock-U to the acid stream (Ru1), >95% recovery proven 1978-99 (Ru1); Ra-226 214-970 Bq/kg vs the 1,000 Bq/kg exemption (Rm11/Rm12); Phase 0 assays at 12 stream points.

Health essentials — the DALY arithmetic and the pilot

\$2,592 per healthy year against the \$4,000/DALY standard; 10 ha spirulina feeds 178,082 children at the RCT dose; basin needs 0.7 ha. Pilot: 100 households + 2 schools, 90 days, \$58,000 — gates: filter uptime $\geq 70\%$, urinary F $\downarrow 20\%$, results public.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

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